

Title: SMS Spam Detection using Ensemble Learning and Boosting

1. Introduction

Spam messages are unwanted messages that may contain advertisements, phishing links, or malicious content. Detecting spam automatically is important for protecting users and improving messaging platforms. Machine Learning techniques can classify messages into Spam or Ham based on text patterns. In this experiment, individual classifiers, ensemble combining methods, and boosting are implemented and compared.

2. Dataset Description

The experiment uses the UCI SMS Spam Collection Dataset.

Total messages: 5574

Classes: Spam and Ham

Ham messages $\approx$ 86%, Spam $\approx$ 14%

The dataset is imbalanced, making recall an important evaluation metric because missing spam messages is risky.

3. Preprocessing

The dataset labels were encoded into numeric values (Spam = 1, Ham = 0).

Text data cannot be directly used in machine learning, so TF-IDF (Term Frequency - Inverse Document Frequency) vectorization was applied. TF-IDF converts messages into numerical feature vectors based on word importance and removes common stopwords.

4. Models Implemented

The following classifiers were implemented:

Base Learners

Naive Bayes (probability-based text classifier)

Logistic Regression (linear classifier)

Linear Support Vector Machine (high-dimensional classifier)

Ensemble Methods

Hard Voting (majority prediction)

Soft Voting (probability averaging)
Stacking (meta-learner = Logistic Regression)

Boosting

AdaBoost using Decision Stumps (Decision Tree with max_depth = 1)

5. Cross Validation

A Stratified 5-Fold Cross Validation was used. The dataset was divided into 5 parts. Each part was used once as testing data while the remaining parts were used for training. Stratification ensures both spam and ham messages appear proportionally in every fold. This provides reliable performance estimation and reduces bias.

6. Evaluation Metrics

The models were evaluated using:

Precision
Recall
F1-Score
ROC-AUC
Confusion Matrix

Recall is important because a false negative (spam predicted as ham) allows harmful messages to reach users.

7. Results and Discussion

The results showed that individual classifiers performed well, especially Linear SVM and Naive Bayes. Voting classifiers improved stability but did not significantly improve recall.

The Stacking classifier achieved the best F1-score, as the meta-learner learned how to combine the strengths of multiple models. Naive Bayes worked well for keyword-based spam, while SVM handled complex patterns, and stacking utilized both.

AdaBoost with decision stumps performed poorly because a depth-1 decision tree can only consider one feature at a time. SMS text classification requires multiple word relationships, which stumps cannot model effectively.

8. Recommendation

Among all combining strategies, **Stacking** is the most suitable approach for SMS spam detection. It performs better than voting because it learns which model to trust rather than simply aggregating predictions. Boosting with decision stumps is less effective for high-dimensional text data, while stacking successfully captures complementary information from multiple classifiers.

9. Conclusion

This experiment demonstrates that ensemble learning improves spam detection performance compared to individual classifiers. Voting provides stability, but stacking gives the highest classification accuracy and balanced precision-recall performance. Therefore, combining classifiers using stacking is the best approach for SMS spam classification on text-based datasets.